Contents

Problem Framing & Data Understanding

What is the problem being solved?

The model is trying to predict whether a customer will pay on their next month payment time using their credit limit, demographic information, bill amounts, and previous payments.

What is the target variable?

The target is to predict of the person will pay at the time or not

- 1 = No, the client
- 0 = Yes, the client

Is it binary or multiclass classification?

It is a **binary classification** problem, because the target has only **two classes,** whether the person will pay at the time or not

Challenges do you expect (e.g., imbalance, noise)

Class Imbalance:

Approximately 78% of the records represent clients who did not default while only 22% represent those who did This skew can lead a model to be overly optimistic, achieving high accuracy by simply predicting "no default" for everyone while failing to identify the high-risk individuals the bank cares about.

Large range between attributes:

The dataset contains variables with **BIG DIFFRENCE** in scales. For instance, credit limits (LIMIT_BAL) and bill amounts reach into the hundreds of thousands, while ages and payment statuses are in double or single digits. Without proper normalization or standardization, the mathematical weight of the larger numbers would dominate the model's learning process.

Categorical Noise:

Columns like EDUCATION and MARRIAGE contain undocumented values (such as 5 or 6) that fall outside the defined data values in the repository. These require careful remapping or grouping to prevent the model from treating "unknown" data as distinct, influential categories.

Data Acquisition & Documentation

Dataset source (link)

https://archive.ics.uci.edu/dataset/350/default+of+credit+card+clients

Any modifications made (cleaning, label changes, training/test splits)

Cleaning

We removed the ID column, as it does not provide any useful information for the model. Additionally, we cleaned the dataset by eliminating noisy records. For example, the marriage attribute contained 54 entries with an invalid value of 0, and the education attribute included inconsistent values such as 0, 5, and 6. Since the dataset is relatively large (30,000 records) and these noisy entries represent only a small fraction (around 1.75%), we chose to remove them to improve data quality

MARRIAGE	
2	15964
1	13659
3	323
0	54

EDUCATION	
2	14030
1	10585
3	4917
5	280
4	123
6	51
0	14

Label changes

We renamed the attributes from X0, X1, X2, …, X23 to more meaningful names based on the repository description. In addition, the monthly delay attributes were inconsistent because they included PAY0 and PAY2 but skipped PAY1, so we corrected the numbering to PAY1, PAY2, …, PAY6.

Training / test splits

We have used K-Fold-cross validation for making the model and we k = 5

Data Cleaning & Preprocessing

Handling missing values

There was no missing value on the data for the categorical or the numeric features based on the repository

Dataset Information

Additional Information

This research aimed at the case of customers' default payments in Taiwan and compares the predictive accuracy of probability of default among six data mining methods. From the perspective of risk management, the result of predictive accuracy of the estimated probability of default will be more valuable than the binary result of classification - credible or not credible clients. ...

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Has Missing Values?

No

Encoding categorical variables

The data (categorical) was already encoded to numbers based on the repository the data like the sex marriage and the educations was mapped to numbers these numbers

Additional Variable Information

This research employed a binary variable, default payment (Yes = 1, No = 0), as the response variable. This study reviewed the literature and used the following 23 variables as explanatory variables:
X1: Amount of the given credit (NT dollar): it includes both the individual consumer credit and his/her family (supplementary) credit.
X2: Gender (1 = male; 2 = female).
X3: Education (1 = graduate school; 2 = university; 3 = high school; 4 = others).
X4: Marital status (1 = married; 2 = single; 3 = others).
X5: Age (year).
X6 - X11: History of past payment. We tracked the past monthly payment records (from April to September, 2005) as follows: X6 = the repayment status in September, 2005; X7 = the repayment status in August, 2005; . . .;X11 = the repayment status in April, 2005. The measurement scale for the repayment status is: -1 = pay duly; 1 = payment delay for one month; 2 = payment delay for two months; . . .; 8 = payment delay for eight months; 9 = payment delay for nine months and above.
X12-X17: Amount of bill statement (NT dollar). X12 = amount of bill statement in September, 2005; X13 = amount of bill statement in August, 2005; . . .; X17 = amount of bill statement in April, 2005.
X18-X23: Amount of previous payment (NT dollar). X18 = amount paid in September, 2005; X19 = amount paid in August, 2005; . . .;X23 = amount paid in April, 2005.

Since we are using the KNN model as a classifier and it relies on distance calculations to assign data points to class 1 or class 2 it can be biased by features with larger numerical values. In categorical features, these numerical differences do not represent any real-world ordering or importance for example being male or female does not imply a higher or lower value. Therefore, we apply one-hot encoding to these features to ensure that all categories are treated equally and do not unfairly influence the distance calculations.

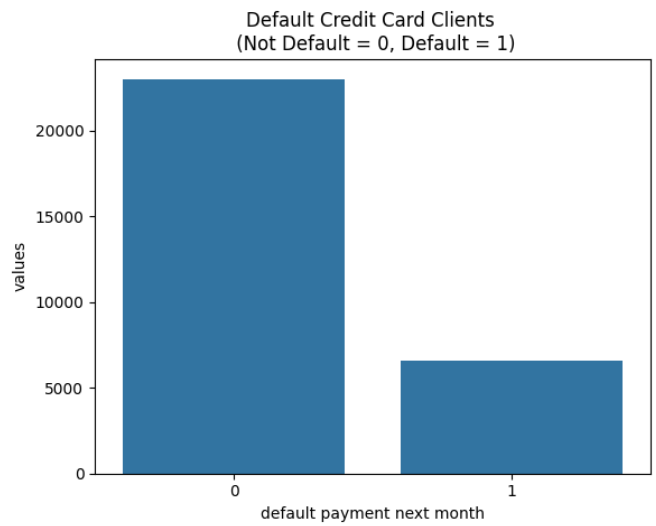
Scaling numerical features (if needed)

We have used Robust scaling for the normalization of the large data and we used it as it don't depend on the outliers (depend on the median of the data) unlike the normalization techniques that rely on the mean of the data which will be smash the small data also we have used it because it is insensitive for the negative value unlike the log transform

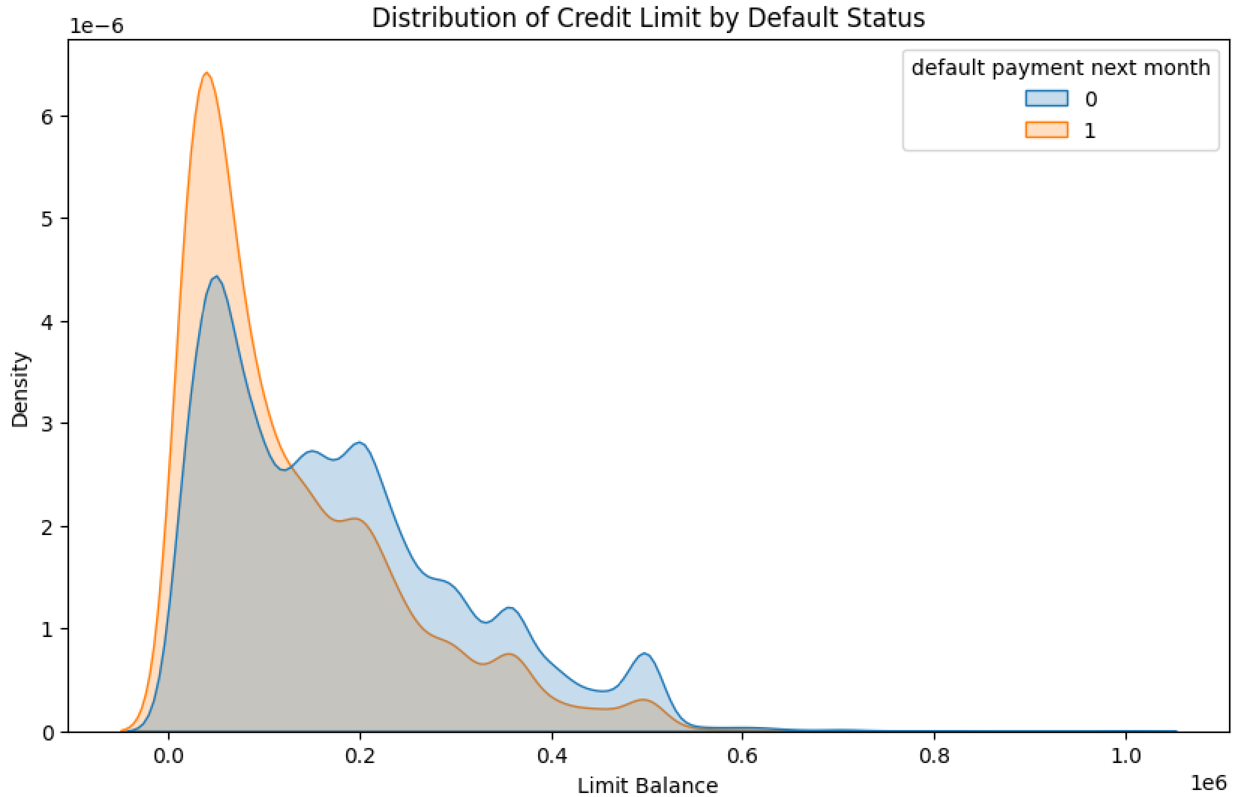
$$x_i = \frac{x - Median}{IQR}$$

Handling class imbalance (if needed)

We used the SMOTE (Synthetic Minority Over-sampling Technique) to oversample the minority positive class and used a sampling_strategy parameter of 0.5 so that after oversampling the positive class would be 0.5 * the size of the negative class decreasing the imbalance between the two classes without altering the dataset too much avoiding corrupted results.

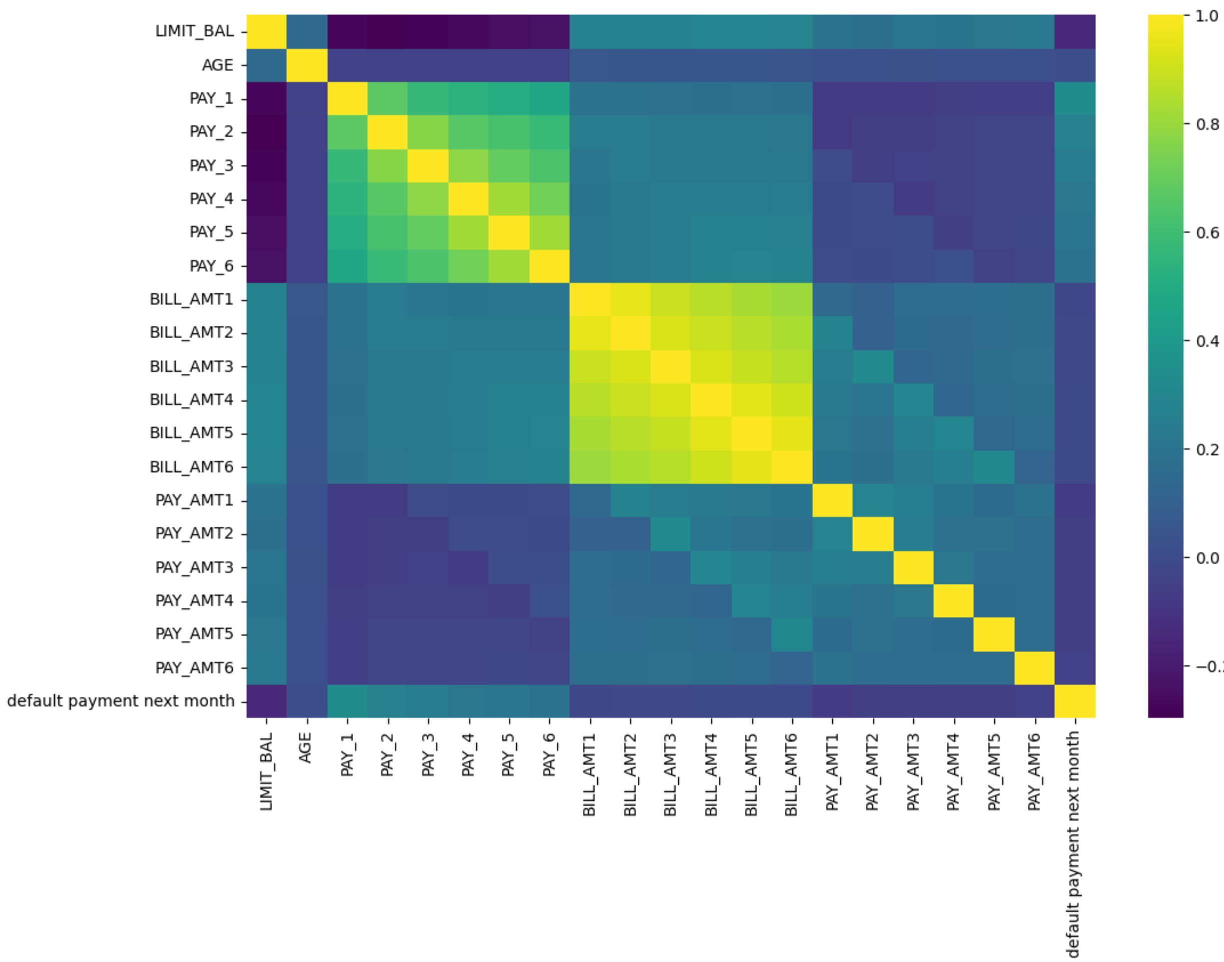


Relation between Limit Balance and output classes



We can see here that by the bank increasing the bill limit the risk of having people who doesn't pay their debts on time increases it's clear how the number of people drops in a general decreasing trend with increasing the limit balance while the number of people who doesn't pay tends to shoots up with increasing limit balance and generally has a slower dropping trend than the number of people who pay in the region between 0.1 -> 0.6.

Heat Map



Conclusions:

- Behavioral Consistency:
  Someone who is late one month is more likely to be late the next month and to have been already late on the prior month.
- Age is not much of a key factor here. Balance limits and payment history are much stronger predictors to the behavior of clients in the future.
- There is a high correlation between the Balance limits amounts on the six months. Same goes for the payment history of the last six months they. All of these carry almost the same information.